Heat & Thermodynamics

Reversible and Irreversible Process: The process in which the system and surroundings can be restored to the initial state from the final state without producing any changes in the thermodynamics properties of the universe is called a **reversible process**.

If heat is produced by the working substance in the direct process, the same quantity will be absorbed by it in the reverse process. Again, if work is done by the substance in the direct process, an equal amount of work will be done on the substance in the reversible process. Thus, it is clear that there is no wastage of energy in the reversible process. In a reversible process, the system passes through a series of equilibrium states. Eg. The slow isothermal and slow adiabatic process.

The process which cannot be retraced in opposite order by reversing the controlling factors is called an **Irreversible process**. Eg.

1. The conduction of heat from a hot to a cold body.
2. Production of heat by friction or by the passage of current through an electrical resistance.
3. Transfer of heat by radiation.
4. Water flows from high level to low level.
5. Current moves from high potential to low potential

Most of the process in nature are irreversible. During an irreversible change the system is not in equilibrium at all instances of time. Irreversible process consists of non-equilibrium states which cannot be represented on a P-V diagram.

2nd Law of Thermodynamics: We have become familiar with different types of energy. All energies provide ability to do work. From first law of thermodynamics we have learnt that heat can be transformed into work and work can be transformed into Heat. But from the first law it is not known in which direction heat will flow or work will be done.

Some special form of 2nd Law of Thermodynamics are stated below-

- ***Clausius's Statement***: It is impossible to transfer heat by an automatic machine from a body at lower temperature to a body at a higher temperature without the help of external energy.
- ***Kelvin's Statement***: Continuous flow of energy cannot be obtained from an object cooling it from its surroundings.
- ***Planck's Statement***: It is impossible to construct an engine that will extract heat continuously from a heat source and will completely transform into work.
- ***Carnot's Statement***: No engine can be built which can extract a fixed amount of heat and will convert it totally into work.

Carnot's Cycle: Sadi Carnot (1832), a French scientist planned an engine which is free from all defects. It is an ideal engine whose efficiency is 100%. It is never possible to give its physical existence. It is only an imaginary engine. In reality, a Carnot's engine works in four steps.

"The cycle in which an ideal gas used as a working substance starts at constant volume, pressure and temperature passes through an isothermal and adiabatic expansion and then through an isothermal and adiabatic compression returns to the initial state is called a Carnot's cycle."

Working of a Carnot's cycle and work done by it can be expressed by a curve. It is called index curve. In the following index curve explanation of different actions and the calculation of work done by a Carnot's cycle has been given.

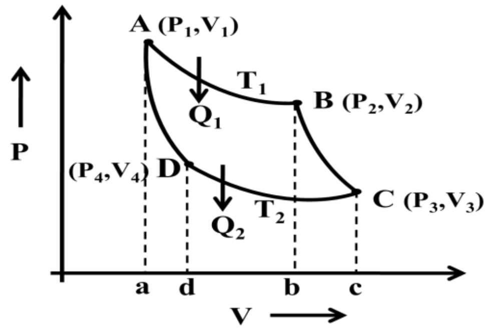

Fig.1: Carnot's cycle index curve.

Step-1 (Reversible isothermal expansion): During this step, the gas is allowed to expand and it does work on the surroundings. The temperature (T_1) of the gas does not change during the process, and thus the expansion is isothermal. The gas expansion is propelled by absorption of heat energy Q_1 from the high temperature reservoir and results in an increase of entropy of the gas.

In index diagram, Work done for the isothermal expansion, $W_1 = area\ ABba$

Step-2 (Reversible adiabatic expansion): For this step, the mechanisms of the engine are assumed to be thermally insulated, thus they neither gain nor lose heat (an adiabatic process). The gas continues to expand, doing work on the surroundings, and losing an amount of internal energy equal to the work that leaves the system. The gas expansion causes it to cool to the "cold" temperature, T_2. The entropy remains unchanged.

In the index diagram, Work done for the adiabatic expansion, $W_2 = area$ BCcb

Step-3 (Reversible isothermal compression): Now the surroundings do work on the gas, causing an amount of heat energy Q_2 to leave the system to the low temperature (constant temperature T_2) reservoir and the entropy of the system decreases.

In index diagram, Work done for the isothermal compression, $W_3 = area\ CcdD$

Step-4 (Reversible adiabatic compression): During this step, the surroundings do work on the gas, increasing its internal energy and compressing it, causing the temperature to rise to T_1 due solely to the work added to the system, but the entropy remains unchanged. At this point the gas is in the same state as at the start of step 1.

In the index diagram, Work done for the adiabatic expansion, $W_4 = area$ BdaA

Now according to convention, work done by the confined gas is positive and work done on this gas is negative. Hence W_1 and W_2 are positive and W_3 and W_4 are negative.

So, total work done by the confined gas is –

$$W = W_1 + W_2 - W_3 - W_4 = area\ ABCD$$

Heat Engine: In order to use the heat energy to work a mechanical system is needed. This mechanical system is called heat engine. That means, 'the engine by which heat energy can be transformed into mechanical energy is called heat engine'. For example – vapour engine, petrol engine, diesel engine etc.

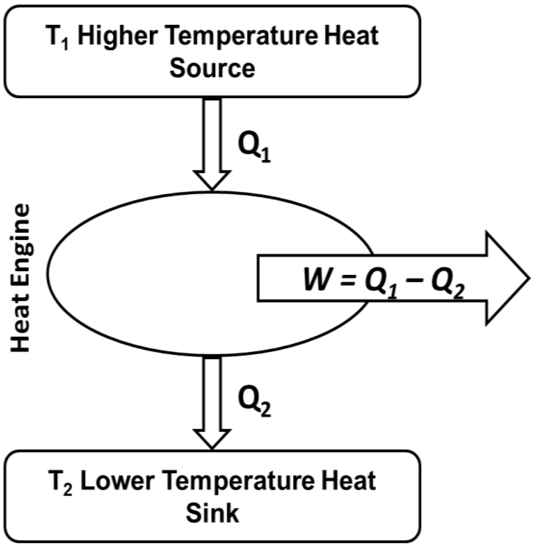

Fig.1: Block diagram of heat engine.

In heat engine, there is one heat source and one heat sink. The engine receives heat from a source and transforms a part of it into work. The amount of heat that is not transformed in to work dissipates in the surroundings and the temperature of the surroundings should be less than that of the engine. That means, engine receives heat from a source of higher temperature, a part of it used as work and the rest amount of heat reject in the heat sink and then the engine returns to its initial state.

Efficiency of Heat Engine: The ratio of the heat energy transformed into work by the engine and the amount of absorbed heat energy by the engine is called the Efficiency of the engine. i.e., Efficiency of the engine,

$$\eta = \frac{Transformed\ heat\ energy\ into\ work\ by\ the\ engine}{Absorbed\ heat\ energy\ by\ the\ engine}$$

$$=> \quad \eta = \frac{W}{Q_1}$$

$$=> \quad \eta = \frac{Q_1 - Q_2}{Q_1}$$

$$=> \quad \eta = 1 - \frac{Q_2}{Q_1}$$

$$\boldsymbol{\eta = \left(1 - \frac{Q_2}{Q_1}\right) \times 100\%}$$

Also, heat and temperature are proportional to each other. So that, $\frac{Q_2}{Q_1} = \frac{T_2}{T_1}$ and hence –

$$\boldsymbol{\eta = \left(1 - \frac{T_2}{T_1}\right) \times 100\%}$$

Problem-1: A heat engine after doing work in each cycle rejects 70% *of heat absorbed from the source, calculate the efficiency of the engine.*

Problem-2: The temperature of the heat source of a Carnot's engine is $400K$*. It takes* $840J$ *heat energy from the source at this temperature and rejects heat energy of* $630J$ *to the cold sink. Find the temperature of the cold sink and efficiency of the engine.*

Problem-3: A Carnot's engine has 50% *efficiency when the sink temperature is* 27^0C*. What is the increase in temperature of the heat source when its efficiency is* 60%*?*

Entropy: In thermodynamics, like temperature, pressure and volume, entropy is also a physical quantity of the system that remain constant in adiabatic process. It is called thermal inertia. Entropy is a measurable quantity. How much change of entropy of a system takes place can be determined.

Entropy of a body changes due to absorption or rejection of heat. Entropy is measured by the rate of change of heat absorbed or rejected with respect to temperature of the system. Suppose a system absorbs or rejects dQ amount of heat at absolute temperature T.

So change of entropy –

$$dS = \frac{dQ}{T}$$

The unit of entropy is JK^{-1}.

- *Entropy in adiabatic process:*

  For adiabatic process, $dQ = 0$, then

$$dS = \frac{dQ}{T} = \frac{0}{T} = 0$$

$$\boldsymbol{S = constant}$$

  The change of entropy is zero in adiabatic process. On the other hand, entropy remain constant in adiabatic process. That is why, another name of the adiabatic process is iso-entropic process.

- *Entropy in reversible and irreversible process:*

  Let us consider a heat engine operated between the source at T_1 and the sink at T_2 temperature. If the amount of heat Q_1 absorbed by the engine from the source and amount of heat Q_2 rejected to the sink, then –

  The increase of entropy for absorption of heat, $S_1 = \frac{Q_1}{T_1}$

  and the decrease of entropy for rejection of heat, $S_2 = \frac{Q_2}{T_2}$

  So that, change in entropy, $dS = S_1 - S_2$

$$dS = \frac{Q_1}{T_1} - \frac{Q_2}{T_2} \dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots (1)$$

  - In case of reversible process, $\frac{Q_1}{T_1} = \frac{Q_2}{T_2}$

Now from equation (1), we get,

$$dS = 0$$

$$\boldsymbol{S = constant}$$

Hence, 'the change in entropy is zero in reversible process'. On the other hand, 'entropy remain constant in reversible process'.

- In case of irreversible process, $\frac{Q_1}{T_1} > \frac{Q_2}{T_2}$

Now from equation (1), we get,

$$\boldsymbol{dS = +ve}$$

Hence, 'the change in entropy is positive in irreversible process'. On the other hand, 'entropy increases in every irreversible process'.

Problem-1: If $4kg$ of water at 100^0C is transformed into vapour of 100^0C. Find the increase in entropy. [Latent heat of vaporization of water $= 2.26 \times 10^6 Jkg^{-1}$]

Problem-2: Find the change in entropy in order to change the temperature of $5kg$ water from 10^0C to 100^0C. [Specific heat of water $= 4.2 \times 10^3 Jkg^{-1}K^{-1}$]

Problem-3: Assignment work suggested by class teacher